1 Gain law

$(e_\theta = \theta - \hat{\theta})$

$$\bar{a}(\bar{w}) = 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)}$$
$$\bar{a}'(\bar{a}, b) := \frac{d\bar{a}}{d\bar{w}} = b\big(1 - \bar{a}\big)^2$$

2 Control law

$(d = 2)$

$$\delta\bar{w}_3[k] = \delta\bar{w}_1[k] + \sum_{i=1}^{d} \Delta\bar{w}_d[k-i] - \sum_{i=1}^{d} \bar{a}_w[k-i]\big\{\bar{f}_{dw}[k-i] + \bar{f}_T[k-i]\big\}$$

$$\boxed{\bar{f}_{dw}[k] = \hat{\bar{a}}_w^{-1}[k]\Big\{\Delta\bar{w}_d[k] + (1-\lambda_c)\sum_{i=1}^{d}\Delta\bar{w}_d[k-i] + (1-\lambda_c)\Big[y_1[k] - \sum_{i=1}^{d}\hat{\bar{a}}_w[k-i]\,\bar{f}_{dw}[k-i]\Big]\Big\}}$$

$$y_1[k] = \delta\bar{w}_1[k] + n_w[k]$$

3 True dynamics

$$\bar{w}[k] = \bar{w}_d[k] - \delta\bar{w}_3[k]$$
$$\Delta\bar{w}[k] := \bar{w}[k+1] - \bar{w}[k]$$
$$e_{\bar{a}_w}[k-i] := \bar{a}_w[k-i] - \hat{\bar{a}}_w[k-i]$$

$$\delta\bar{w}_3[k+1] = \delta\bar{w}_3[k] + \Delta\bar{w}_d[k] - \bar{a}_w[k]\big\{\bar{f}_{dw}[k] + \bar{f}_T[k]\big\} \tag{1}$$

$$\bar{a}_w[k]\,\bar{f}_{dw}[k] = \hat{\bar{a}}_w[k]\,\bar{f}_{dw}[k] + e_{\bar{a}_w}[k]\,\bar{f}_{dw}[k] \tag{2}$$

$$\begin{aligned}\hat{\bar{a}}_w[k]\,\bar{f}_{dw}[k] = {} & \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + (1-\lambda_c)\sum_{i=1}^{d} e_{\bar{a}_w}[k-i]\,\bar{f}_{dw}[k-i] \\ & + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]\end{aligned} \tag{3}$$

$$\delta\bar{w}_3[k+1] = \lambda_c\,\delta\bar{w}_3[k] - \bar{f}_{dw}[k]\,e_{\bar{a}_w}[k] - (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i]\,e_{\bar{a}_w}[k-i] - \varepsilon_w[k] \tag{4}$$

$$\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]$$
$$\bar{a}_w[k+1] = \bar{a}_w[k] + b\big(1-\bar{a}_w[k]\big)^2\,\Delta\bar{w}[k]$$
$$b[k+1] = b[k]$$

4 Estimator

$$\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b\,\big]^\top$$

$$\delta\hat{\bar{w}}_1[k+1] = \delta\hat{\bar{w}}_2[k]$$
$$\delta\hat{\bar{w}}_2[k+1] = \delta\hat{\bar{w}}_3[k]$$
$$\delta\hat{\bar{w}}_3[k+1] = \lambda_c\,\delta\hat{\bar{w}}_3[k]$$
$$\Delta\hat{\bar{w}}[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]$$
$$\hat{\bar{a}}_w[k+1] = \hat{\bar{a}}_w[k] + \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\,\Delta\hat{\bar{w}}[k]$$
$$\hat{b}[k+1] = \hat{b}[k]$$

5 Error dynamics

(True − Estimator; coefficients at the estimate)

$$e_{\Delta\bar{w}}[k] := \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k]$$

$$e_{\delta\bar{w}_1}[k+1] = e_{\delta\bar{w}_2}[k]$$
$$e_{\delta\bar{w}_2}[k+1] = e_{\delta\bar{w}_3}[k]$$
$$e_{\delta\bar{w}_3}[k+1] = e_{\delta\bar{w}_3}[k] - e_{\Delta\bar{w}}[k] \tag{5}$$
$$e_{\bar{a}_w}[k+1] = \big(1-2\hat{b}(1-\hat{\bar{a}}_w)\Delta\hat{\bar{w}}[k]\big)e_{\bar{a}_w}[k] + \hat{b}(1-\hat{\bar{a}}_w)^2 e_{\Delta\bar{w}}[k] + (1-\hat{\bar{a}}_w)^2\Delta\hat{\bar{w}}[k]\,e_b[k] \tag{6}$$
$$e_b[k+1] = e_b[k]$$

$$e_{\Delta\bar{w}}[k] = (1-\lambda_c)\,e_{\delta\bar{w}_3}[k] + \bar{f}_{dw}[k]\,e_{\bar{a}_w}[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i]\,e_{\bar{a}_w}[k-i] + \varepsilon_w[k] \tag{7}$$

$$
\begin{aligned}
e_{\Delta\bar{w}}[k-1] &= e_{\delta\bar{w}_2}[k] - e_{\delta\bar{w}_3}[k] \\
e_{\Delta\bar{w}}[k-2] &= e_{\delta\bar{w}_1}[k] - e_{\delta\bar{w}_2}[k]
\end{aligned}
$$

$$
e_{\bar{a}_w}[k-1] = e_{\bar{a}_w}[k] - \hat{b}(1-\hat{\bar{a}}_w)^2\big(e_{\delta\bar{w}_2}[k] - e_{\delta\bar{w}_3}[k]\big) - (1-\hat{\bar{a}}_w)^2\Delta\hat{\bar{w}}[k-1]\,e_b[k] + \mathcal{O}\big(\Delta\hat{\bar{w}}\,e\big) \tag{8}
$$

$$
\begin{aligned}
e_{\bar{a}_w}[k-2] = e_{\bar{a}_w}[k] &- \hat{b}(1-\hat{\bar{a}}_w)^2\big(e_{\delta\bar{w}_1}[k] - e_{\delta\bar{w}_3}[k]\big) \\
&- (1-\hat{\bar{a}}_w)^2\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)e_b[k] + \mathcal{O}\big(\Delta\hat{\bar{w}}\,e\big)
\end{aligned} \tag{9}
$$

$$
\begin{aligned}
e_{\Delta\bar{w}}[k] = (1-\lambda_c)\,e_{\delta\bar{w}_3}[k] &+ F_{dw}[k]\,e_{\bar{a}_w}[k] \\
&- (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2\Big(\bar{f}_{dw}[k-1]\big(e_{\delta\bar{w}_2}[k] - e_{\delta\bar{w}_3}[k]\big) + \bar{f}_{dw}[k-2]\big(e_{\delta\bar{w}_1}[k] - e_{\delta\bar{w}_3}[k]\big)\Big) \\
&- (1-\lambda_c)(1-\hat{\bar{a}}_w)^2\Big(\bar{f}_{dw}[k-1]\,\Delta\hat{\bar{w}}[k-1] + \bar{f}_{dw}[k-2]\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)\Big)e_b[k] \\
&+ \varepsilon_w[k]
\end{aligned} \tag{10}
$$

$$
F_{dw}[k] = \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i]
$$

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2\bar{f}_{dw}[k-2]\,e_{\delta\bar{w}_1}[k] \\
&\quad + (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2\bar{f}_{dw}[k-1]\,e_{\delta\bar{w}_2}[k] \\
&\quad + \Big[\lambda_c - (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2\big(\bar{f}_{dw}[k-1] + \bar{f}_{dw}[k-2]\big)\Big]e_{\delta\bar{w}_3}[k] \\
&\quad - F_{dw}[k]\,e_{\bar{a}_w}[k] \\
&\quad + (1-\lambda_c)(1-\hat{\bar{a}}_w)^2\Big(\bar{f}_{dw}[k-1]\,\Delta\hat{\bar{w}}[k-1] \\
&\qquad + \bar{f}_{dw}[k-2]\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)\Big)e_b[k] \\
&\quad - \varepsilon_w[k]
\end{aligned}
$$

$$
\begin{aligned}
e_{\bar{a}_w}[k+1] &= -(1-\lambda_c)\,\hat{b}^2(1-\hat{\bar{a}}_w)^4\bar{f}_{dw}[k-2]\,e_{\delta\bar{w}_1}[k] \\
&\quad - (1-\lambda_c)\,\hat{b}^2(1-\hat{\bar{a}}_w)^4\bar{f}_{dw}[k-1]\,e_{\delta\bar{w}_2}[k] \\
&\quad + (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2\Big[1 + \hat{b}(1-\hat{\bar{a}}_w)^2\big(\bar{f}_{dw}[k-1] + \bar{f}_{dw}[k-2]\big)\Big]e_{\delta\bar{w}_3}[k] \\
&\quad + \Big[1 + \hat{b}(1-\hat{\bar{a}}_w)^2F_{dw}[k] - 2\hat{b}(1-\hat{\bar{a}}_w)\Delta\hat{\bar{w}}[k]\Big]e_{\bar{a}_w}[k] \\
&\quad + \Big[(1-\hat{\bar{a}}_w)^2\Delta\hat{\bar{w}}[k] \\
&\qquad - (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^4\Big(\bar{f}_{dw}[k-1]\,\Delta\hat{\bar{w}}[k-1] \\
&\qquad\quad + \bar{f}_{dw}[k-2]\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)\Big)\Big]e_b[k] \\
&\quad + \hat{b}(1-\hat{\bar{a}}_w)^2\varepsilon_w[k]
\end{aligned}
$$

$$
e_b[k+1] = e_b[k]
$$

$$
F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
(1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2 \times \bar{f}_{dw}[k-2] & (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2 \times \bar{f}_{dw}[k-1] & \lambda_c - (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2 \times \big(\bar{f}_{dw}[k-1] + \bar{f}_{dw}[k-2]\big) & -F_{dw} & (1-\lambda_c)(1-\hat{\bar{a}}_w)^2 \times \Big(\bar{f}_{dw}[k-1]\,\Delta\hat{\bar{w}}[k-1] + \bar{f}_{dw}[k-2]\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)\Big) \\
-(1-\lambda_c)\,\hat{b}^2(1-\hat{\bar{a}}_w)^4 \times \bar{f}_{dw}[k-2] & -(1-\lambda_c)\,\hat{b}^2(1-\hat{\bar{a}}_w)^4 \times \bar{f}_{dw}[k-1] & (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^2 \times \Big(1 + \hat{b}(1-\hat{\bar{a}}_w)^2 \times \big(\bar{f}_{dw}[k-1] + \bar{f}_{dw}[k-2]\big)\Big) & 1 + \hat{b}(1-\hat{\bar{a}}_w)^2 F_{dw} - 2\hat{b}(1-\hat{\bar{a}}_w)\Delta\hat{\bar{w}} & (1-\hat{\bar{a}}_w)^2\Delta\hat{\bar{w}} - (1-\lambda_c)\,\hat{b}(1-\hat{\bar{a}}_w)^4 \times \Big(\bar{f}_{dw}[k-1]\,\Delta\hat{\bar{w}}[k-1] + \bar{f}_{dw}[k-2]\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)\Big) \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
\mathbf{q}[k] = \Big[\, 0,\ 0,\ -\varepsilon_w[k],\ \hat{b}(1-\hat{\bar{a}}_w)^2\varepsilon_w[k],\ 0 \,\Big]^\top
$$

6 Measurement

$\big(\nabla_d \bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]\big)$

$$
\begin{aligned}
y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k], \qquad \bar{a}_w[k-d] = \bar{a}_w[k] - b\big(1-\bar{a}_w[k]\big)^2 \nabla_d \bar{w}_d[k] \\
\hat{y}_1[k] &= \delta \hat{\bar{w}}_1[k] \\
\hat{y}_2[k] &= \hat{\bar{a}}_w[k] - \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2 \nabla_d \bar{w}_d[k] \\
H_{24}[k] &= \frac{\partial \hat{y}_2[k]}{\partial \hat{\bar{a}}_w[k]} = 1 + 2\,\hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)\,\nabla_d \bar{w}_d[k] \\
H_{25}[k] &= \frac{\partial \hat{y}_2[k]}{\partial \hat{b}[k]} = -\big(1-\hat{\bar{a}}_w[k]\big)^2 \nabla_d \bar{w}_d[k] \\
H[k] &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{24}[k] & H_{25}[k] \end{bmatrix} \\
\mathbf{e}_y[k] &= H[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}
\end{aligned}
$$

7 Noise

$$
\begin{aligned}
Q[k] &= Q_{33}[k]\,\mathbf{g}[k]\,\mathbf{g}[k]^\top, \qquad \mathbf{g}[k] = \Big[\,0,\ 0,\ -1,\ \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2,\ 0\,\Big]^\top, \qquad Q_{55} = 0 \\
Q_{33}[k] &= \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \frac{4k_B T\,a_o}{R} \\
R[k] &= \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\bar{a}}_w[k]\big)\big(\hat{\bar{a}}_w[k] + \xi\big)^2
\end{aligned}
$$